EXTENDING A TENSOR FUNCTOR FROM COHERENT TO QUASI-COHERENT MODULES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4495-4503 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4505-4572, the complete author proof (66 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_stacks-stacks-more-morphisms.json (pb=2989, stmt 4495-4503, proof 4505-4572).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: additivity/tensor compatibility omissions are naturality bookkeeping - the load-bearing checks are in-span.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-extend' also exists in stacks-properties, stacks-spaces-limits, stacks-topologies — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 002P: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4572–4572. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FWY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4572–4572. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GR6: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4572–4572. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GRF: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4572–4572. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GQZ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4572–4572. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Noetherian algebraic stacks. Any right exact tensor functor $F : Coh(\mathcal{O}_\mathcal{X}) \to Coh(\mathcal{O}_\mathcal{Y})$ extends uniquely to a right exact tensor functor $F : QCoh(\mathcal{O}_\mathcal{X}) \to QCoh(\mathcal{O}_\mathcal{Y})$ commuting with all colimits.*

Proof. The existence and uniqueness of the extension is a general fact, see Categories, Lemma 0FWY. To see that the lemma applies observe that coherent modules on locally Noetherian algebraic stacks are by definition modules of finite presentation, see Cohomology of Stacks, Definition 0GR6. Hence a coherent module on $\mathcal{X}$ is a categorically compact object of $QCoh(\mathcal{O}_\mathcal{X})$ by Cohomology of Stacks, Lemma 0GQZ. Finally, every quasi-coherent module is a filtered colimit of its coherent submodules by Cohomology of Stacks, Lemma 0GRF.

Since F is additive, also the extension of F is additive (details omitted). Since F is a tensor functor and since colimits of modules commute with taking tensor products, also the extension of F is a tensor functor (details omitted).

In this paragraph we show the extension commutes with arbitrary direct sums. If $\mathcal{F} = \bigoplus_{j\in J} \mathcal{H}_j$ with $\mathcal{H}_j$ quasi-coherent, then $\mathcal{F} = \operatorname{colim}_{J'\subset J \text{ finite}} \bigoplus_{j\in J'} \mathcal{H}_j$. Denoting the extension of F also by F we obtain

$$\begin{aligned} F(\mathcal{F}) &= \operatorname{colim}_{J'\subset J \text{ finite}} F(\bigoplus\nolimits_{j\in J'} \mathcal{H}_j) \\ &= \operatorname{colim}_{J'\subset J \text{ finite}} \bigoplus\nolimits_{j\in J'} F(\mathcal{H}_j) \\ &= \bigoplus\nolimits_{j\in J} F(\mathcal{H}_j) \end{aligned}$$

Thus F commutes with arbitrary direct sums.

In this paragraph we show that the extension is right exact. Suppose $0 \to \mathcal{F} \to \mathcal{F}' \to \mathcal{F}'' \to 0$ is a short exact sequence of quasi-coherent $\mathcal{O}_\mathcal{X}$-modules. Then we write $\mathcal{F}' = \bigcup \mathcal{F}'_i$ as the union of its coherent submodules (see reference given above). Denote $\mathcal{F}''_i \subset \mathcal{F}''$ the image of $\mathcal{F}'_i$ and denote $\mathcal{F}_i = \mathcal{F} \cap \mathcal{F}'_i = \operatorname{Ker}(\mathcal{F}'_i \to \mathcal{F}''_i)$. Then it is clear that $\mathcal{F} = \bigcup \mathcal{F}_i$ and $\mathcal{F}'' = \bigcup \mathcal{F}''_i$ and that we have short exact sequences

$$0 \to \mathcal{F}_i \to \mathcal{F}'_i \to \mathcal{F}''_i \to 0$$

Since the extension commutes with filtered colimits we have $F(\mathcal{F}) = \operatorname{colim}_{i\in I} F(\mathcal{F}_i)$, $F(\mathcal{F}') = \operatorname{colim}_{i\in I} F(\mathcal{F}'_i)$, and $F(\mathcal{F}'') = \operatorname{colim}_{i\in I} F(\mathcal{F}''_i)$. Since filtered colimits of sheaves of modules is exact we conclude that the extension of F is right exact.

The proof is finished as a right exact functor which commutes with all coproducts commutes with all colimits, see Categories, Lemma 002P. □